

MAKSATIKHA, Russian Federation

WMO: 272080

Lat: 57.800N

Lon: 35.883E

Elev: 142

StdP: 99.63

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-22.4	-28.2	0.3	-25.4	-24.9	0.4	-22.2	6.5	-3.9	6.1	-3.7	0.8	60	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.2	20.0	27.0	19.2	24.9	18.0	21.1	27.5	20.0	25.3	19.0	23.5	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.5	18.0	13.2	22.7	17.1	12.5	21.5	61.7	27.8	57.9	25.2	54.6	23.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.3	4.7	DB	-30.4	31.2	4.3	2.3	-33.5	32.8	-36.0	34.2	-38.4	35.5	-41.5	37.1	(4)
(5)			WB	-30.5	22.2	4.3	1.2	-33.6	23.0	-36.1	23.7	-38.5	24.3	-41.6	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-8.3	-7.7	-2.5	4.7	12.4	15.3	18.5	16.2	11.0	4.9	-0.4	-3.9	(6)	
	DBStd	10.40	7.43	7.05	5.23	4.15	4.52	3.59	3.31	3.68	3.37	4.32	4.74	5.99	(7)	
	HDD10.0	2583	567	495	387	167	26	3	0	2	28	164	311	432	(8)	
	HDD18.3	4921	826	728	646	410	192	105	38	87	222	417	561	691	(9)	
	CDD10.0	786	0	0	0	7	99	161	263	194	57	6	0	0	(10)	
	CDD18.3	83	0	0	0	0	7	13	42	21	1	0	0	0	(11)	
	CDH23.3	817	0	0	0	0	82	121	402	209	4	0	0	0	(12)	
	CDH26.7	222	0	0	0	0	17	23	119	62	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.5	2.3	2.5	2.2	2.2	2.1	1.8	1.9	2.1	2.3	2.6	2.7	(14)	
(15) Precipitation	PrecAvg	643	46	38	31	35	56	74	74	69	54	63	55	50	(15)	
	PrecMax	821	80	69	46	68	126	167	173	167	123	110	123	85	(16)	
	PrecMin	468	13	8	10	11	14	3	6	1	10	15	3	26	(17)	
	PrecStd	100	17	14	11	15	27	34	47	34	28	27	28	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	3.5	12.4	21.1	29.1	29.6	33.1	32.3	23.3	17.4	10.0	7.8	(19)	
		MCWB	3.7	1.7	6.1	10.5	19.2	19.4	21.4	22.2	16.7	12.5	8.8	7.1	(20)	
	2%	DB	2.6	2.2	8.5	17.7	25.5	26.7	29.8	28.0	20.7	13.4	7.8	5.8	(21)	
		MCWB	1.9	0.9	4.0	9.5	16.7	18.7	20.6	19.6	15.9	10.9	6.5	5.0	(22)	
	5%	DB	1.6	1.4	5.7	14.7	23.0	24.3	27.4	25.4	18.7	12.1	6.5	4.0	(23)	
		MCWB	0.7	0.4	2.8	8.2	15.3	17.4	19.6	19.1	14.0	10.4	5.5	3.1	(24)	
	10%	DB	0.6	0.7	3.9	12.3	20.5	22.3	25.2	23.1	16.9	10.9	5.3	2.3	(25)	
		MCWB	0.0	-0.1	1.8	7.1	13.5	16.3	18.3	17.6	13.4	9.5	4.5	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	2.2	7.0	11.7	20.7	21.1	22.2	22.6	18.3	13.2	9.2	7.1	(27)	
		MCDB	4.5	3.0	12.5	17.9	28.2	26.7	30.1	28.7	21.5	15.1	9.6	7.7	(28)	
	2%	WB	2.0	1.2	4.6	10.4	18.0	19.8	21.1	21.0	16.1	11.7	7.0	5.0	(29)	
		MCDB	2.4	1.8	7.1	16.0	23.5	25.5	28.1	26.8	19.5	12.9	7.6	5.7	(30)	
	5%	WB	0.9	0.6	3.2	9.1	16.1	18.4	20.2	19.7	14.9	10.6	5.6	3.1	(31)	
		MCDB	1.3	1.2	5.3	14.1	21.0	22.5	25.7	24.1	18.1	12.0	6.3	3.9	(32)	
	10%	WB	0.0	-0.1	2.1	7.5	14.7	17.1	19.3	18.3	13.7	9.5	4.5	1.7	(33)	
		MCDB	0.7	0.6	3.8	11.9	19.3	21.0	24.1	22.2	16.6	10.9	5.2	2.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.4	7.7	9.7	10.7	9.9	10.2	10.0	8.4	5.9	3.7	4.1	(35)	
		MCDBR	3.4	4.3	9.7	14.7	14.6	12.6	13.5	13.9	12.0	8.5	4.9	4.1	(36)	
	5% WB	MCWBR	3.5	3.7	6.3	7.9	6.7	5.9	5.5	6.5	7.0	5.9	4.4	4.1	(37)	
		MCWBR	3.2	3.9	8.4	12.9	12.7	10.9	11.5	12.1	10.2	7.0	4.3	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.318	0.336	0.386	0.377	0.366	0.401	0.399	0.362	0.340	0.311	0.270	(40)	
		taud	2.104	2.075	2.139	2.211	2.363	2.412	2.330	2.324	2.403	2.379	2.224	2.098	(41)	
	Ebn at Noon		549	689	799	818	858	872	828	800	777	684	515	440	(42)	
		Edh at Noon	54	88	109	120	111	107	114	107	85	66	49	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.44	1.22	2.72	3.97	5.18	5.37	5.25	4.13	2.55	1.12	0.39	0.23	(44)	
	RadStd		0.07	0.21	0.33	0.43	0.42	0.47	0.66	0.36	0.35	0.21	0.06	0.06	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page